AA Pro

Functional Requirements Document

FRD v1.1 | Children’s AI Learning Portal

13 October 2026

Updated requirements baseline with visual UML-style use cases, component architecture, logical ERD, sequence diagrams, functional/non-functional requirements, security, operations and live-readiness criteria.

Document control

Field	Value
Version	1.1
Date	13 October 2026
Approved brand	AA Pro
Status	Requirements baseline; implementation/integrations are not assumed to exist
Change from v1.0	Owner decisions consolidated; clear rendered diagrams and detailed use cases added.

Important interpretation

This document defines intended behaviour and proposed design. It is not proof that code, provider accounts, permissions or integrations have been implemented. Diagrams are logical models subject to engineering validation.

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1. Product overview and scope

AA Pro is a responsive web portal for children, parents/guardians, approved human teachers and platform administrators. It combines AI-class scheduling, parent-managed profiles and learning content, human-teacher scheduling, an ElevenLabs voice classroom, textbook/PDF viewer, Zoom meetings and notifications.

1.1 Goals

- Students schedule AI classes in permitted time; students cannot manage human-teacher bookings.
- Display voice AI alongside authorised textbook/PDF content.
- Parents manage children, uploads, assignments and human-teacher bookings.
- Teachers manage only their approved calendar and assigned students/resources.
- Super Admin manages Parent/Teacher accounts with audited actions.
- Protect child data through server-side authorisation, minimisation and retention controls.

1.2 Initial out of scope

- Payments/subscription billing; Google Classroom integration; embedded Zoom; native mobile apps; school SIS replacement; raw audio retention by default.

2. Actors, roles and permissions

Capability	Student	Parent	Approved Teacher	Super Admin	External invite
Schedule AI	Yes	For linked child	Not initially	Support only	No
Book human meeting	No	Yes	Authorised student only	Support only	No portal access
Join human meeting	If invited	If invited	If invited/host	Only if authorised	Email invitation link
Upload content	No	Yes	Share assigned resources	Admin controls	No
Manage child	No	Yes	No	Account admin only	No
Manage users	No	No	No	Yes	No
Access unrelated child	No	No	No	Only as necessary and audited	No

All permissions are enforced server-side on every API/resource operation. Hiding a UI control is not authorisation. Super Admin access is least-privilege and audited.

3. Use cases and use-case diagram

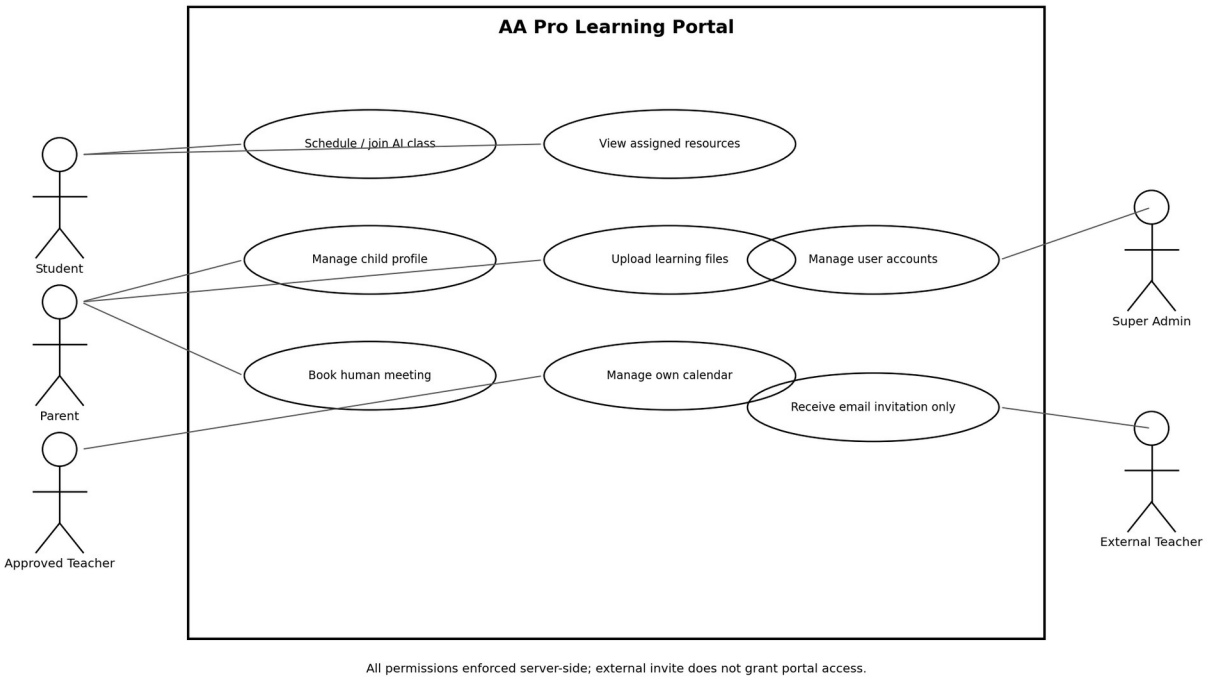


Figure 1. High-level UML use-case view.

3.1 Primary use cases

ID / use case	Actor	Precondition	Outcome
UC-01 Sign in	All registered roles	Active account and valid credentials	Authenticated role-bound session
UC-02 Manage child profile	Parent	Authenticated parent	Child profile and subjects maintained

UC-03 Schedule AI class	Student/Parent	Active child and permitted slot	AI booking created without conflict
UC-04 Join AI class	Student	Authorised booking/resources	Secure AI session starts
UC-05 Upload content	Parent	Valid file and permissions	Private resource stored/assigned
UC-06 Book human meeting	Parent	Linked child, eligible teacher/slot	Confirmed only after required provider success
UC-07 Manage calendar	Approved teacher	Approved account	Own availability and authorised meetings managed
UC-08 Invite external teacher	Parent	Meeting/invitation flow	Email invitation only; no portal access
UC-09 Receive reminders	Authorised recipients	Active booking and configured channel	Attempt recorded; status visible
UC-10 Manage accounts	Super Admin	Elevated account and MFA/policy as designed	Parent/Teacher account changes audited

3.2 Use-case rules

- Student schedules AI only; cannot create/cancel/reschedule human meetings.
- Parent sees and manages only linked children.
- External teacher invite does not create account or permissions.
- Approved teacher accesses own calendar and assigned students only.
- Super Admin actions on accounts and sensitive records are audited.

4. Functional requirements

FR-001 Authentication

- Secure sign-in/out, recovery, expiry, secure cookies, rate limits and deactivation.
- Parent creates child credentials; child login is linked to parent/guardian.

FR-002 Roles and relationships

- Roles: Student, Parent, Teacher, Super Admin.
- Reject cross-child/cross-teacher access and audit role/relationship changes.

FR-003 Child profiles

- Parent maintains grade, curriculum, timezone and subjects.
- Initial subjects: Maths, Science, SST, English, Hindi and IT; configurable catalog.

FR-004 Teacher onboarding

- Approved onboarding required for teacher calendar/resource access.
- External invitation does not create a portal account.

FR-005 Availability

- AI available 24/7 except blocked periods, existing bookings and parent restrictions.
- Teacher slots include time, timezone, recurrence, capacity and eligibility.
- No double booking; availability changes cannot silently remove confirmed bookings.

FR-006 Student AI booking

- Select subject, optional chapter, date and permitted slot.
- Backend rechecks role, blocks and conflicts at confirmation; use idempotency key.

FR-007 Parent AI booking

- Parent selects linked child, subject, optional chapter and slot.

- Parent may view/cancel/reschedule under configured AI booking rules.

FR-008 Human booking

- Parent books approved teacher or initiates an external teacher invite.
- Students cannot book/cancel/reschedule human meetings.
- Provider failure must not leave booking falsely confirmed.

FR-009 Zoom

- Phase 1: Basic/free for evaluation; 40-minute maximum scheduled human meeting.
- Join action opens Zoom externally; embedding deferred.
- Use real provider-created meeting ID/join URL only. Verify API, host permissions and concurrency before production.

FR-010 AI classroom

- Secure ElevenLabs session initiation; show connection/microphone/speaking states and End Class.
- Handle microphone denial, network loss, timeout, reconnect and provider errors.

FR-011 Textbook viewer

- Show authorised PDF alongside AI; responsive stacked layout on mobile.
- Agent page changes require validated structured tool/event and resource/page allowlist.

FR-012 Parent uploads

- Configure supported formats/size limits; validate type/signature; scan where feasible.
- Private object storage; assign by child/subject/chapter; handle corrupt/duplicate files.

FR-013 Google Classroom

- Not in initial release; reconsider if parent demand justifies it.
- If approved later, read-only first; validate OAuth, admin policy, quotas and revocation.

FR-014 Learning library

- Browse authorised resources by subject/chapter/type/source.
- Distinguish parent uploads, teacher resources and future external sources.

FR-015 Notifications

- Parent-configurable email/WhatsApp; email initial fallback.
- Default T−15 reminder subject to provider, recipient and consent.
- Durable jobs, retries, idempotency and stale-job suppression.

FR-016 Session history

- Separate booking from actual AI session outcome.
- Store summaries and progress; raw audio disabled by default.

FR-017 Super Admin

- Add/edit/deactivate Parent and Teacher accounts.
- Audit sensitive changes; no integration secret exposure.

FR-018 Audit/operations

- Log booking, upload, security, integration and admin events.
- Correlation IDs and alerts; avoid secrets and unnecessary child content.

FR-019 Timezone

- Store UTC instants and user IANA timezone; display per user.
- Handle daylight-saving where applicable.

FR-020 External invitation/join

- External invite sends email only, no portal access.
- Join redirects to actual Zoom URL and exposes it only to authorised participants.

5. End-to-end workflows and UML sequence diagrams

Sequence diagrams are logical models; confirm details against selected frameworks and provider APIs.

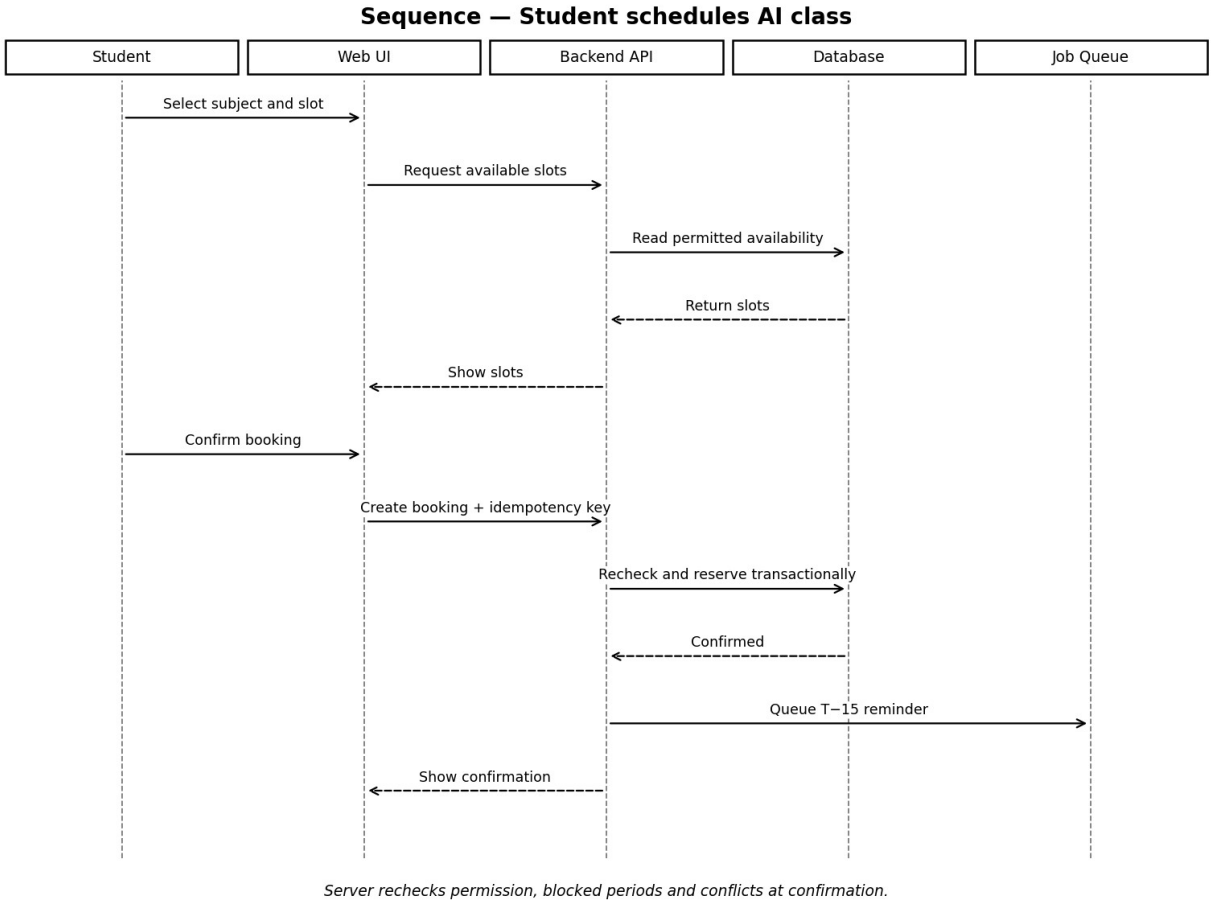


Figure 2. Student AI-class booking sequence.

5.1 Student schedules AI class

1. Student selects subject, optional chapter, date and permitted slot.
2. Backend validates role, blocked periods, parent restrictions and conflicts.
3. Backend transactionally rechecks and reserves slot with idempotency key.
4. Queue reminder for start minus 15 minutes according to preferences.
5. At class time, validate booking/resource access and start secure AI session.
6. Save allowed session metadata and summary/progress on completion.

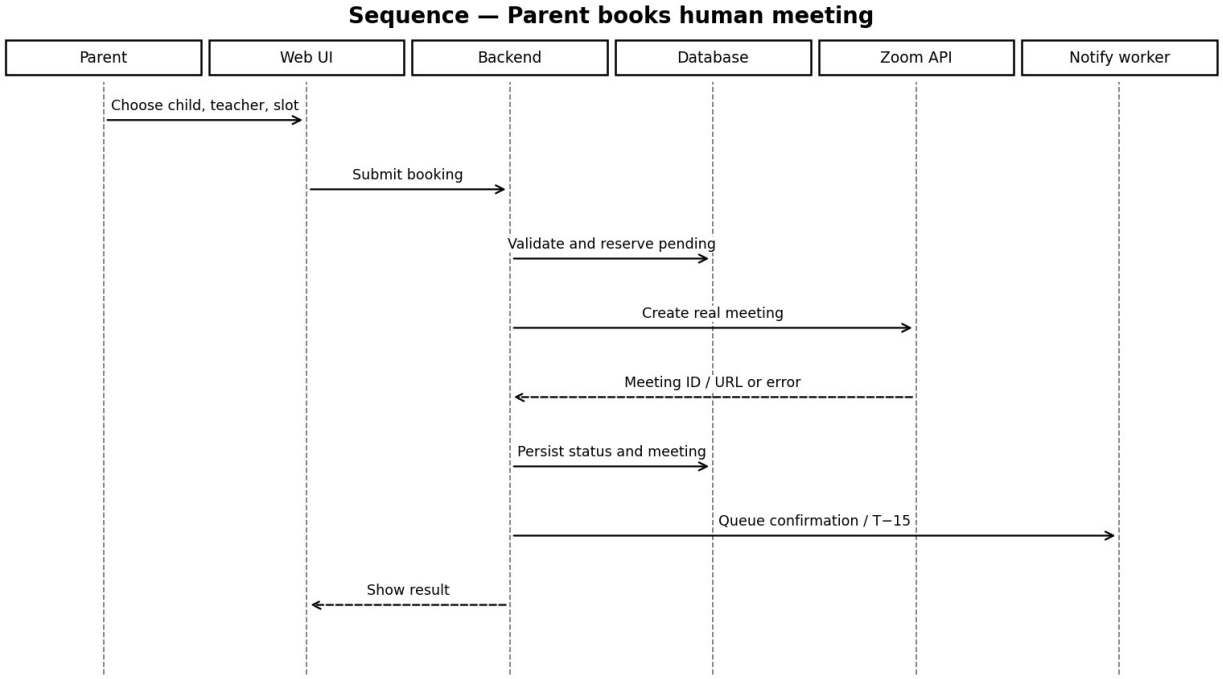


Figure 3. Parent books human meeting and Zoom creation.

5.2 Parent books human meeting

7. Parent chooses linked child, teacher and slot.
8. Backend validates relationship, availability, 40-minute limit and conflicts.
9. Create pending reservation and request Zoom meeting creation if account/app supports it.
10. Confirm only after provider returns real meeting ID/join URL; otherwise show actionable failure.
11. Queue confirmation and T−15 reminder. Join opens Zoom externally.

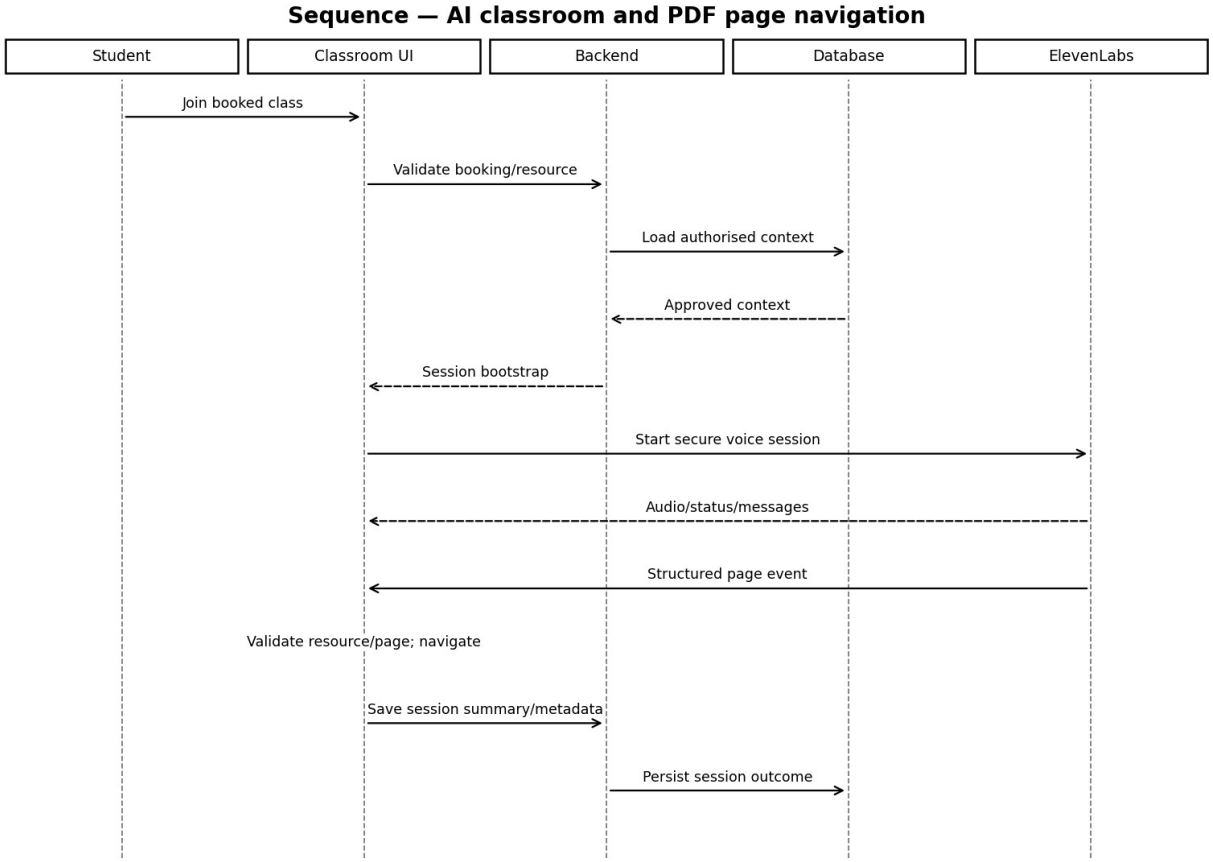


Figure 4. AI classroom and structured PDF page navigation.

5.3 AI class and textbook synchronisation

12. Validate booking/resource; load authorised context.
13. Render textbook/PDF beside voice agent.
14. Structured page-navigation event is checked against resource and page bounds.
15. End session and persist permitted metadata/summary.

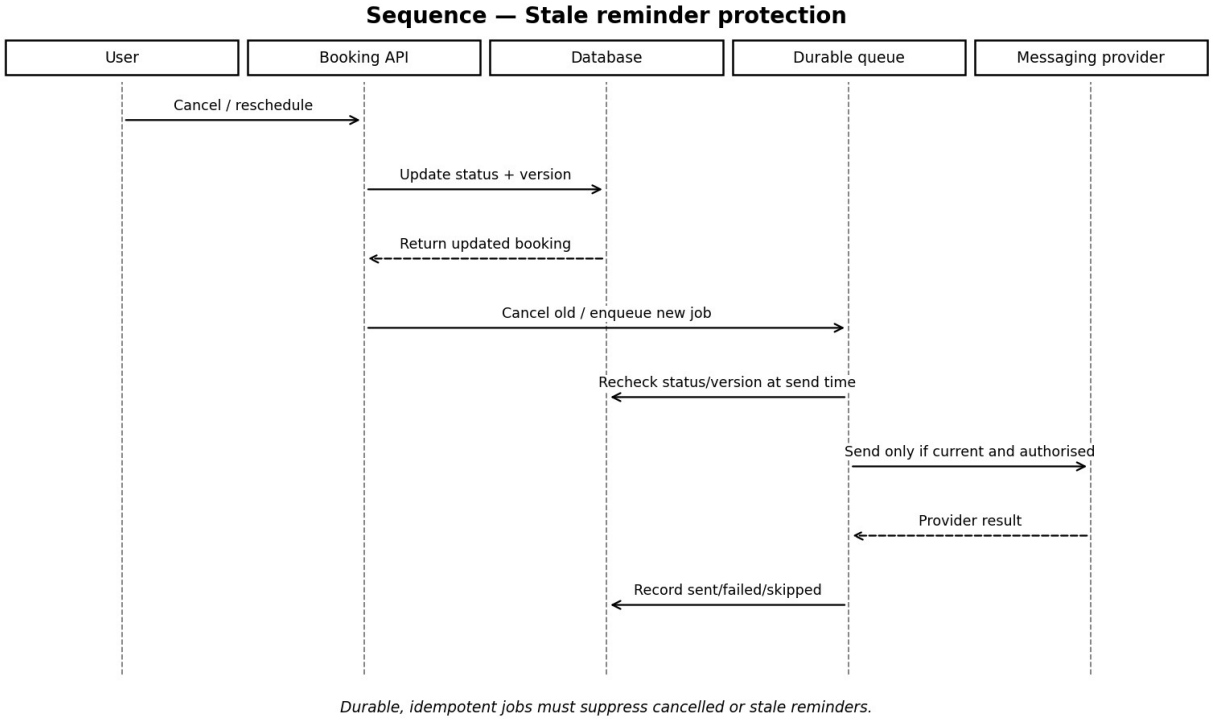


Figure 5. Cancellation/rescheduling and stale reminder protection.

5.4 Reminder lifecycle

16. Cancellation/reschedule increments booking version and updates status.
17. Cancel old jobs and enqueue replacement reminder.
18. Worker rechecks booking status/version immediately before sending.
19. Record sent/failed/skipped result; suppress stale jobs.

6. State models

Object	States	Control
Booking	PENDING_PROVIDER, CONFIRMED, RESCHEDULE_PENDING, CANCELLED, COMPLETED, MISSED, FAILED	Confirm only after provider success where required.
AI session	NOT_STARTED, CONNECTING, IN_PROGRESS, ENDED, DISCONNECTED, FAILED	Separate from booking state.
Notification	QUEUED, SENDING, SENT, DELIVERED, FAILED, RETRY_PENDING, SKIPPED	SENT is not DELIVERED.
Integration	NOT_CONNECTED, CONNECTED, DEGRADED, EXPIRED, REVOKED, ERROR	Never expose tokens.
Resource	UPLOADING, PROCESSING, AVAILABLE, FAILED, ARCHIVED	Only AVAILABLE resources open.
Teacher	INVITED, PENDING_APPROVAL, ACTIVE, SUSPENDED, DEACTIVATED	Invite does not grant portal access.

7. Logical data model and database requirements

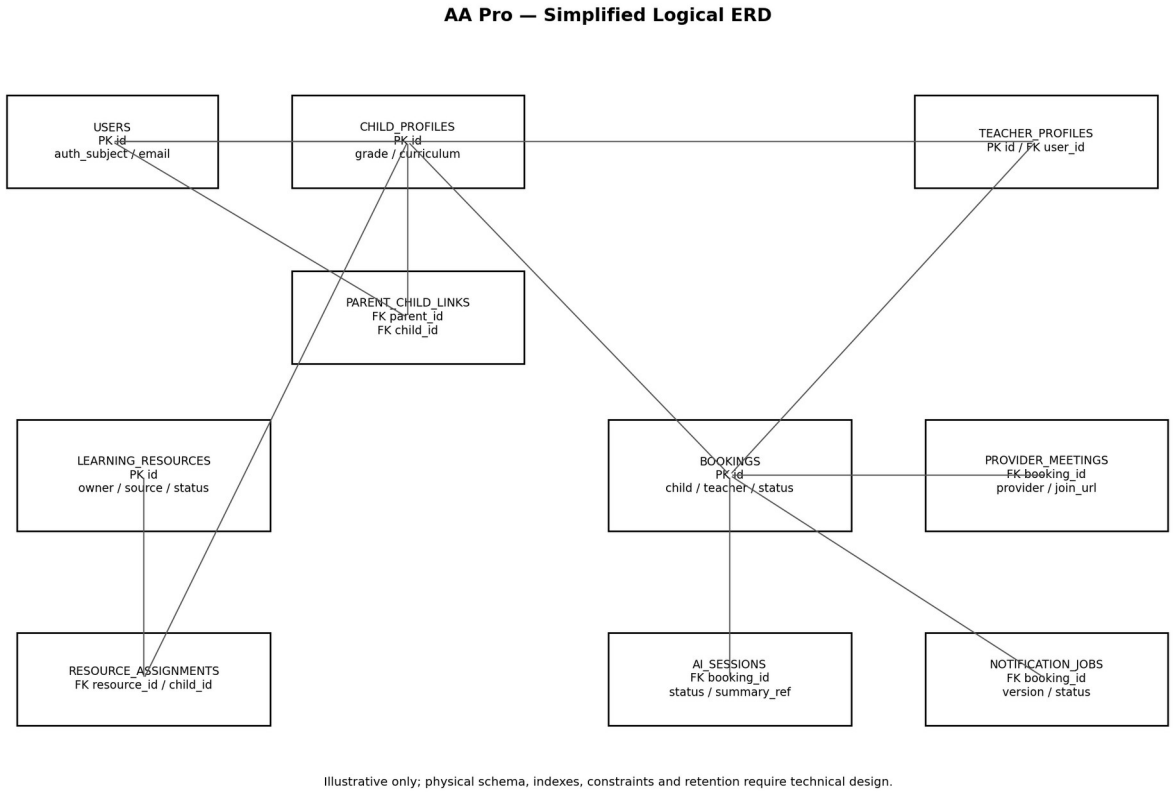


Figure 6. Simplified logical ERD; not the final physical schema.

Entity	Illustrative fields / purpose
users / user_roles	Identity, auth subject, nullable email, roles, status.
parent_child_links / child_profiles	Guardian relationship, grade, curriculum, timezone.
teacher_profiles / teacher_student_links	Approved teacher profile and assigned student relationships.
subjects	Configurable subject catalogue.
availability_rules / slots	Recurrence, owner, UTC start/end, timezone, capacity and status.
bookings / participants	Session type, child, parent, teacher, times, status, version and authorised participants.
provider_meetings	Provider, external meeting ID, join URL and status.
learning_resources / assignments	Owner, source, storage key, file type, child assignment and visibility.
ai_sessions / learning_progress	Conversation reference, timestamps, status, summary and progress.
notification_preferences / jobs / attempts	Channel, consent, destination reference, schedule, idempotency and provider status.
audit_events	Actor, action, entity, timestamp and correlation ID.

7.1 Database integrity

- PostgreSQL or equivalent relational database; private object storage for files.
- Foreign keys, migrations, backups and restore procedures.
- Transactions/locking/constraints prevent overlapping confirmed bookings.
- Indexes for child/time/status, teacher/time/status and slot/status.
- Idempotency keys for booking creation and notifications.
- UTC instants + IANA timezone; retention/deletion policy required before production.
- Server-side relationship checks; row-level security as defence in depth.

8. Logical architecture and deployment

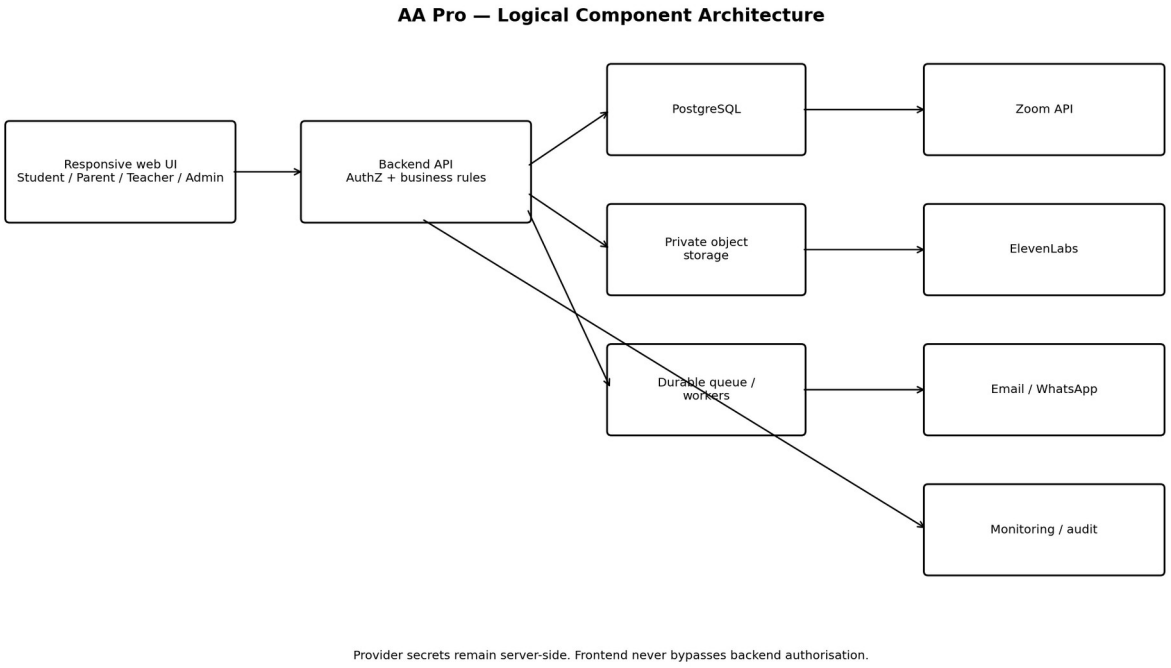


Figure 7. Logical component architecture.

- Responsive web frontend for all four registered roles.
- Backend API is authoritative for access, booking and provider operations.
- PostgreSQL for structured records; private object storage for files.
- Durable worker/queue for reminders and retries; never rely on in-memory timers.
- Secrets/OAuth tokens stay server-side; structured logs, metrics and alerts.

9. Non-functional requirements

ID	Category	Initial target / requirement
NFR-01	Availability	99.5% monthly target excluding declared provider outages; confirm hosting SLA.
NFR-02	Performance	p95 <2s common API reads under agreed test load.
NFR-03	Responsive UX	Primary flows usable on desktop/tablet/mobile without horizontal overflow.
NFR-04	Voice	Benchmark first AI response and set acceptance threshold.
NFR-05	Booking integrity	No duplicate confirmed booking for an exclusive slot.
NFR-06	Reminder integrity	T−15 default; recheck booking version/status before send.
NFR-07	Security	TLS, least privilege, secret storage, server authz, rate limits, dependency scans.
NFR-08	Privacy	Minimise child data; consent, retention and deletion policy before production.
NFR-09	Accessibility	Target WCAG 2.2 AA for key flows where feasible.
NFR-10	Reliability	Bounded retries, idempotency, graceful provider failure and reconciliation.
NFR-11	Observability	Structured logs, correlation IDs, alerts and

		redaction.
NFR-12	Maintainability	Typed interfaces, modular provider adapters, migrations and automated tests.
NFR-13	Scalability	Start small; load-test before expansion; avoid premature microservices.
NFR-14	Recovery	Automated backups and restore drill; define RPO/RTO.
NFR-15	Compatibility	Define and test supported browser matrix.
NFR-16	Data residency	Select region after privacy, latency, provider and legal review.
NFR-17	File safety	Type/signature validation, limits, private buckets, scanning where feasible.
NFR-18	Auditability	Record changes without secrets or unnecessary conversation content.

10. Hardware, hosting and environments

- Modern browser device and stable internet; microphone/audio for AI; camera only if Zoom video is used.
- No dedicated on-prem hardware for cloud-hosted web app.
- Hosting needs: web runtime, managed PostgreSQL, private object storage, durable queue/worker, email provider, Zoom and ElevenLabs.
- Separate local, test, staging and production; synthetic child data in non-production.

11. Integrations and external dependencies

Integration	Prerequisites and risk
ElevenLabs	Agent/SDK and secure session initiation; handle microphone, device variance, latency, quotas and cost.
Zoom	Verify Basic/API, host permissions and concurrency; external launch; 40-minute maximum; never fabricate join URL.
Email	Verified sender/domain, API credentials, bounce/error tracking.
WhatsApp	Enable after business/provider setup, consent and applicable templates.
File storage	Private object storage, signed access, file validation and scanning where feasible.
Google Classroom	Out of initial scope; if approved later, read-only first and validate OAuth/admin restrictions.

12. Security, privacy and child safety

- Threat-model account takeover, cross-child access, exposed meeting links, malicious files, token theft and AI tool misuse.
- Enforce server-side checks on every API/resource route; private storage and expiring links.
- Keep secrets/tokens server-side; validate AI tool parameters and allowlist actions.
- Define guardian consent, privacy notice, retention/deletion and legal review before production.
- No raw audio by default; avoid unrelated model training on child data unless authorised and legally permitted.
- Rate limiting, secret scanning, dependency updates, incident response and audit logs.

13. Operations and support

- Monitor availability, latency, errors, DB health, queue depth, reminders and provider errors.
- Alert on persistent job failures, token revocation, backup failures and suspicious access denial.
- Document secret rotation, provider outage, rollback, support and restore procedures.

- Track voice, messaging, Zoom, hosting, storage and egress cost.

14. Updated owner decisions and technical verification

ID	Approved decision / status
OD-01	Parent manages family and creates child profile/credentials; child has individual login; no human meeting management or file upload.
OD-02	Approved onboarding required for teacher portal/calendar/resources; external invite grants email only.
OD-03	AI available 24/7 except blocked periods, conflicts and parent restrictions.
OD-04	Students cannot cancel/reschedule human meetings.
OD-05	Zoom Basic/free for evaluation; 40-minute max; external launch; verify API/host/concurrency; embedding deferred.
OD-06	Google Classroom not in initial scope.
OD-07	Parent-configurable email/WhatsApp; email initial fallback.
OD-08	Store session summaries and progress; raw audio disabled by default.
OD-09	Payments out of initial scope; free subscriptions are later-phase concept only.
OD-10	Super Admin manages Parent/Teacher accounts; audit sensitive actions.
OD-11	UTC storage and each user's configured IANA timezone.
OD-12	External Zoom launch first; embedded Zoom deferred.
TV-01	Verify Zoom Basic/API host permissions and account model.
TV-02	Verify concurrent meeting/licensed-host rules.
TV-03	Configure email; enable WhatsApp only after provider/consent readiness.
TV-04	Implement safe child credential recovery, lockout and deactivation.

15. Acceptance criteria and release gates

Gate	Exit criteria
G1 Design	Canonical Stitch screens, AA Pro branding and role flows approved.
G2 Identity	Role/relationship checks and cross-child negative tests pass.
G3 Scheduling	AI scheduling works; human student actions rejected; 40-minute max enforced; concurrency tests pass.
G4 AI classroom	Authorised voice session/PDF works; microphone/error states and structured page navigation tested.
G5 Notifications	T−15 jobs, cancellation/reschedule suppression and duplicate prevention tested.
G6 Zoom	Real meeting ID/URL; external join; API/host permissions and failure paths tested.
G7 Teacher access	External invite has no portal access; approved teacher access is scoped.
G8 Live readiness	Privacy/security review, monitoring, backup/restore, cost review, rollback and owner approval.

16. Risk register

Risk	Impact	Mitigation
Zoom Basic/API/host model unsuitable	High	Prove early; retain external redirect; do not assume concurrency.
Permission leakage	Critical	Server policies, relationship checks and negative tests.
Double booking	High	Transactions, constraints and idempotency.
Stale reminders	High	Durable jobs recheck status/version before sending.

Zoom creation failure	High	Pending/failed state; never falsely confirm.
Voice/browser incompatibility	Medium/High	Browser tests and fallback/error states.
Unsafe uploads	High	Type/size/signature checks, private storage and scanning.
Child privacy/consent unclear	Critical	Minimisation, guardian controls and legal review.
Provider cost/quota	Medium/High	Usage budgets, alerts, backoff and quotas.

17. Recommended implementation sequence

20. Audit Stitch ZIP and approve canonical screens/roles.
21. Create requirements traceability matrix.
22. Build responsive UI with mock data for all roles.
23. Implement authentication, relationships and server-side authorisation.
24. Implement availability, booking constraints and state models.
25. Implement private uploads/library.
26. Integrate ElevenLabs/PDF and structured page navigation.
27. Prove Zoom API/host permissions and external join in development.
28. Implement durable email reminders; add WhatsApp after readiness.
29. Add monitoring, backups, restore tests, security review and controlled rollout.

18. Definition of live-ready

AA Pro is live-ready only when scope is implemented and tested; permissions are server-enforced; double booking is prevented; integrations are authorised; reminders respect cancellation/rescheduling; child data is protected; monitoring and backups are active; recovery/support are documented; Zoom account permissions are verified; and the product owner approves production. A design prototype or successful API connection alone is not live readiness.

Appendix: Official technical references

- ElevenLabs React SDK — https://elevenlabs.io/docs/eleven-agents/libraries/react
- Google Classroom API (future scope) — https://developers.google.com/workspace/classroom/reference/rest
- Zoom Meetings API — https://developers.zoom.us/docs/api/meetings/
- Zoom Meeting SDK for web (future embedded option) — https://developers.zoom.us/docs/meeting-sdk/web/